

FDK™ Toolkit, Finance Domain Fairness Audit

External Validation Report: German Credit Data (UCI Statlog) & HMDA (Washington State, 2025)

Third Real-World Benchmark Validation, Following Justice/COMPAS, Education/ Bar Passage

Dataset: German Credit and HMDA (Washington 2025)

Date: 22 July 2026

Abstract

This report documents the real-dataset validation of the FDK™ (Fairness Diagnostic Kit) Finance domain following a no-code user-experience redesign of its column-mapping workflow. Validation was conducted against two independent real-world credit-decision datasets: the UCI Statlog German Credit dataset and a Washington State extract of Home Mortgage Disclosure Act (HMDA) data. Three product-affecting bugs were identified and fixed during this process, and the Finance confirmation-page UX was redesigned around the needs of non-technical, no-code users rather than data scientists. Both datasets required a derived (toy classifier) prediction column, as neither contains a native model-decision field. The audits themselves surfaced materially different fairness profiles across datasets and group cuts: German Credit showed high bias on both tested cuts (personal_status_sex, foreign_worker); HMDA showed high bias on the race cut but low bias on the sex cut, on the same underlying loan population, underscoring that fairness findings do not generalize across protected attributes even within one dataset.

Background

Following the FDK Toolkit's original seven-domain bug remediation and two prior real-dataset validations (Justice/COMPAS and Education/Bar Passage), the project moved to Finance as the third domain for real-dataset validation. This surfaced a product-design problem rather than a pure bug: the manual column-override dropdowns, originally built for the Education domain, displayed every column in an uploaded dataset with no filtering, unusable for FDK's actual target audience of no-code users auditing their own data, as distinct from data scientists already familiar with tools such as AIF360. This reframed the remaining Finance work around building a reference no-code UX that the other six domains could later adopt.

Dataset Descriptions

German Credit Data (UCI Statlog)

- Source: UCI Machine Learning Repository, Statlog (German Credit Data) dataset
- Used the categorical german.data file (not german.data-numeric, which one-hot-encodes every categorical field and destroys the readable group columns this validation depends on)
- Preprocessing: added UCI-standard column headers; decoded personal_status_sex and foreign_worker from raw numeric codes to readable labels; recoded the target from {1,2} to {0,1} as credit_risk_good
- n = 1,000 applicants, 21 columns

Derived prediction column: German Credit contains no native model-decision field, confirmed exhaustively, with only one binary {0,1} column in the entire dataset (the target itself). A toy LogisticRegression classifier was trained (features = all other columns, one-hot encoded, intentionally including protected attributes as inputs, matching how real-world credit models are commonly built) to generate predicted_credit_risk and predicted_probability. Training accuracy was 78.6%, with genuine off-diagonal disagreement in the confusion matrix confirming this is not a trivial or memorized fit.

HMDA (Home Mortgage Disclosure Act), Washington State Extract

- Source: CFPB HMDA Data Browser (<https://ffiec.cfpb.gov/data-browser/data/>), filtered to Washington State only, no institution (LEI) or ethnicity pre-filter
- Raw extract: 322,182 records, activity year 2025

Decision filtering: action_taken was restricted to the three codes representing genuine underwriting decisions, 1 (Originated), 2 (Approved but not accepted), 3 (Denied), excluding withdrawn applications, incomplete files, loan purchases, and preapproval-only requests, which are not accept/deny decisions and would corrupt a naive binary mapping. This left 230,123 decisioned applications. loan_approved was derived as 1 for codes {1,2}, 0 for code 3.

Derived prediction column: as with German Credit, HMDA contains no native model-decision field. A toy LogisticRegression classifier (StandardScaler-normalized numeric features; one-hot encoded categorical features including derived_race, derived_ethnicity, and derived_sex as inputs) was trained on the full decisioned set. Test accuracy was 84.85%, with genuine off-diagonal disagreement (35,003 of 230,123 rows) confirming a non-trivial fit.

Group-cut cleaning: each protected-attribute column was cleaned of non-single-applicant categories ("Not Available", "Joint", "Free Form Text Only") before use as a group cut, per the two-cut selection made for this validation:

- Race cut (derived_race): n = 180,056, 6 groups (White, Asian, Black or African American, American Indian or Alaska Native, Native Hawaiian or Other Pacific Islander, 2 or more minority races)
- Sex cut (derived_sex): n = 113,160, 2 groups (Male, Female)

Findings: No-Code UX Redesign & Bug Remediation

UX Redesign

Finance uses a two-step flow, unlike Education's single-step flow: an upload page posts to a confirmation page, which is where the audit's column mapping is actually finalized before the audit runs. The confirmation page (auto_confirm_finance.html) was identified as the correct home for no-code-friendly UX, and was redesigned so that all four mapping fields (group, target, prediction, probability) are real dropdowns, each pre-populated from a filtered, role-appropriate candidate list, pre-selected to the auto-detected suggestion, and annotated with a plain-language description and a domain-specific naming-convention hint (e.g. "in finance datasets, this is often called default, approved, or loan_status"). Where no valid prediction-column candidate exists, the dropdown is replaced with an explicit warning rather than silently defaulting to an incorrect column.

Finding 1, Probability column excluded from its own dropdown

Root cause: the identifier-exclusion check in the candidate-filtering logic (`n_unique == len(series)`) fired on genuine continuous probability columns, since every value in a real probability score is naturally distinct, the same test that correctly identifies ID columns as unusable also incorrectly flagged a valid probability column.

Evidence: confirmed via direct pandas inspection of a real probability column before the fix was proposed.

Remediation: the identifier-exclusion check was scoped to apply only to group/target/prediction roles, never to the probability role.

Verification: confirmed the probability column now appears correctly in its dropdown on subsequent real-dataset uploads.

Finding 2, Upload-page target dropdown unfiltered (deferred, not fixed)

The upload page's own target dropdown pulls raw, unfiltered columns from a shared cross-domain endpoint used by all seven FDK domains. This surfaced clearly on the 80+ column HMDA dataset. Filtering it properly requires either extending that shared endpoint to return per-role filtered candidates, or treating the upload-page dropdown as a low-stakes preview with the confirmation page as the real, filtered decision point. This is a deliberate cross-domain design decision, flagged for the wider FDK roadmap rather than patched in isolation; see Scope Limitations.

Finding 3, Non-binary column silently corrupting a report

Root cause: a non-binary column (`dependents_count`, values `{1,2}`) was accepted as a prediction column via a fallback check that tested only `nunique() == 2` rather than genuine `{0,1}` membership.

Evidence: confirmed by direct inspection of a corrupted report showing `selection_rate` values above 1.0 and negative `predicted_negatives_per_group` counts, both mathematically impossible for a genuine binary prediction column.

Remediation: fixed at three layers, (1) the detection fallback now requires `set(...).issubset({0,1})`; (2) the confirmation page's own sanitization step nulls out any suggested mapping that isn't in its role's valid-candidate list before it is stored in session; (3) the pipeline's metric-calculation step independently rejects non-binary or object-dtype target/prediction columns as a last line of defense.

Verification: confirmed no further malformed reports were produced across all subsequent real-dataset runs in this validation.

Fairness Findings

German Credit, `personal_status_sex`

Group	n	Selection rate	Base rate
male_married_widowed	92	0.8587	0.7283
male_single	548	0.8084	0.7336
female_divorced/separated/married	310	0.6871	0.6484
male_divorced/separated	50	0.6600	0.6000

- Statistical parity difference: 0.1987
- Disparate impact: 0.7686 (below the conventional 0.80 fair-lending threshold)
- Composite bias score: 0.1857, overall assessment: HIGH_BIAS
- FPR difference: 0.3930; FNR difference: 0.1204 (largest error-rate gap observed across both datasets)

German Credit, `foreign_worker`

Group	n	Selection rate	Base rate
no	37	0.9730	0.8919
yes	963	0.7601	0.6926

Thin-subgroup caveat: the "no" (non-foreign-worker) group is only $n=37$, above the project's 20-sample floor, but small enough that its metrics should be read as indicative rather than statistically robust.

- Statistical parity difference: 0.2128
- Disparate impact: 0.7812
- Composite bias score: 0.1699, overall assessment: HIGH_BIAS

HMDA, derived_race

Group	n	Selection rate	Base rate
White	142,178	0.9199	0.8140
Asian	25,914	0.9055	0.8115
American Indian or Alaska Native	2,188	0.8455	0.7358
Black or African American	7,747	0.8287	0.7261
Native Hawaiian or Other Pacific Islander	1,232	0.7938	0.7037
2 or more minority races	797	0.7616	0.6964

Thin-subgroup caveat: "2 or more minority races" (n=797) is the smallest group here, comfortably above the 20-sample floor but still the least statistically robust cut in this table.

- Statistical parity difference: 0.1583 (White selection rate 0.9199 vs. "2 or more minority races" 0.7616)
- Disparate impact: 0.8280, clears the 0.80 fair-lending bright line, but narrowly
- Composite bias score: 0.1203, overall assessment: HIGH_BIAS
- FPR difference: 0.1901; FNR difference: 0.0933

HMDA, derived_sex

Group	n	Selection rate	Base rate
Male	69,680	0.8818	0.7725
Female	43,480	0.8763	0.7683

- Statistical parity difference: 0.0055
- Disparate impact: 0.9937, comfortably clear of the 0.80 threshold
- Composite bias score: 0.0042, overall assessment: LOW_BIAS

Cross-dataset observations

The same HMDA loan population that showed HIGH_BIAS on the race cut (composite bias score 0.1203, disparate impact 0.8280) showed LOW_BIAS on the sex cut (composite bias score 0.0042, disparate impact 0.9937). This is a genuine, verified contrast, not an artifact of dataset choice, and demonstrates that a fairness audit's conclusion is specific to the protected attribute tested: auditing only one group cut on a given dataset does not license a general conclusion about that dataset's fairness. Both German Credit cuts (personal_status_sex, foreign_worker) returned HIGH_BIAS, though on a much smaller sample (n=1,000) than HMDA (n=180,056 / 113,160), which affects how much weight the two datasets' findings should carry relative to one another.

Scope Limitations

- Neither dataset contains a native model-decision column; both required a toy LogisticRegression classifier to derive predicted outcomes. These fairness findings describe the toy models' behavior, not any real deployed lending model; they validate that FDK's pipeline correctly detects and quantifies disparities, not that any specific real-world lender exhibits these disparities.

- The upload page's target dropdown remains unfiltered (Finding 2), a deliberate, deferred cross-domain design decision, not an oversight.
- Several subgroups across both datasets are thin (German Credit foreign_worker "no", n=37; HMDA "2 or more minority races", n=797), all clear the project's 20-sample statistical-power floor, but findings on these groups specifically should be read as indicative rather than robust.
- HMDA's toy classifier produced predicted_probability values at the extremes of the 0–1 range on some rows, typical of logistic regression fit against strongly predictive categorical combinations; this does not affect the binary prediction column used for the core fairness metrics but is noted for completeness.
- This validation covers one U.S. state (Washington) for HMDA; findings should not be generalized to other states or to national HMDA patterns without further testing.

Conclusion

The FDK Finance domain's no-code UX redesign and three associated bug fixes are verified working across two independent real-world credit-decision datasets. The validation confirms both that the pipeline correctly computes and reports fairness metrics against genuine data, and that a single dataset can yield materially different fairness conclusions depending on which protected attribute is audited, reinforcing that FDK's multi-group-cut capability is a functional necessity, not a redundant feature. Finance is now established as the reference implementation for the remaining six FDK domains' no-code UX work.